Design and ASIC Implementation of a Decision-Making System for Bone Fracture Detection

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Abstract*— Accurate identification of bone fractures from X-ray images is of prime importance for their timely orthopaedic treatment, particularly in high-pressure or resource-constrained clinical scenarios. Most traditional software-based image-processing pipelines and deep-learning models are afflicted with high computational overhead, dependence on bulky annotated datasets, noise susceptibility, and limited amenability to real-time embedded deployment. This work presents a new fully hardware-accelerated decision-making system for detecting bone fractures, which is implemented as a custom ASIC. The proposed architecture integrates a streaming 3×3 Scharr gradient extraction module, adaptive direction-aware binary masking, and a rule-based scoring engine that assesses cluster density, orthogonal corner patterns, chaotic edge behaviour, and localised brightness variation. A dedicated majority-vote decision block combines these multi-directional cues to obtain the final diagnosis in hardware. Unlike most conventional software-centric designs, the proposed ASIC system offers deterministic timing, low power consumption, and high structural interpretability, making it well suited for portable and point-of-care medical platforms. Experimental evaluation on real X-ray images performed in this paper demonstrates the high diagnostic reliability of this architecture.

***Keywords*— ASIC Design, Bone Fracture Detection, Scharr Operator, VLSI Architecture, Medical Image Processing, Decision-Making Hardware, Binary Mask Classification, Low-Power Systems, Edge Detection**.

I. INTRODUCTION

Fractures are among the most common orthopaedic injuries and require early diagnosis to prevent long-term complications. Although X-ray imaging is widely used, subtle fractures, low contrast, and high clinical workload make manual interpretation challenging and prone to error. This has led to increasing interest in automated fracture-detection systems for decision support. Most existing approaches rely on software-based image processing or deep-learning models, which demand high computational resources and large datasets, limiting their suitability for real-time, low-power embedded applications. In contrast, dedicated VLSI architectures provide compact, energy-efficient, and deterministic solutions for point-of-care diagnosis.

This work presents an ASIC-based fracture detection system incorporating Scharr gradient extraction, adaptive edge classification, and rule-based decision making. Implemented in 45-nm CMOS technology, the design achieves 77 MHz operation, 1.151 mW power consumption, and an area of 33,047.460 μm². The fully pipelined architecture processes 256×256 images in under 1 ms, enabling throughput exceeding 1000 frames per second. Experimental results on X-ray images demonstrate 85% detection accuracy, highlighting the potential of VLSI-based solutions for portable medical diagnostics.

II. RELATED WORK

Automatic bone fracture detection has been explored using classical image processing, FPGA-based accelerators, and deep learning approaches. Early FPGA-based work by Abdullah and Younis [1] demonstrated real-time edge analysis but showed limited robustness under varying imaging conditions. The Darkroom framework [2] introduced key hardware principles such as streaming architectures and pipelined processing, which inspired later ASIC-based designs. Edge detection remains central to fracture identification, with FPGA implementations of Sobel, Prewitt, and Canny operators achieving high throughput through parallelism [4]–[7], though they often rely on external software for diagnostic interpretation.

Software-based approaches using MATLAB employ segmentation, thresholding, and morphological operations to detect fractures [9]–[10], but are computationally intensive and unsuitable for real-time embedded systems. Similarly, methods based on feature extraction and machine learning [3] achieve moderate accuracy (~83%) but lack hardware efficiency, while Canny–Hough-based pipelines [4] are sensitive to noise and computationally expensive. Deep learning techniques [14], [15] provide high accuracy but require large datasets and significant computational resources, limiting their applicability in low-power portable systems.

In contrast, the proposed work presents a complete ASIC-based fracture-detection pipeline that integrates Scharr

gradient extraction, adaptive masking, and rule-based decision-making. The design achieves 77 MHz operation, 1.151 mW power, and 85% detection accuracy, offering a deterministic, low-power alternative suitable for real-time point-of-care diagnostic applications.

III. PROPOSED ARCHITECTURE

The proposed system is a fully hardware-accelerated pipeline for real-time bone fracture detection using rule-based analysis of X-ray images. It processes one pixel per clock cycle and produces a final diagnosis per frame. The architecture consists of three stages:

(1) Window Formation

(2) Gradient and Mask Computation

(3) Rule-Based Decision Making.

A. Architectural Overview

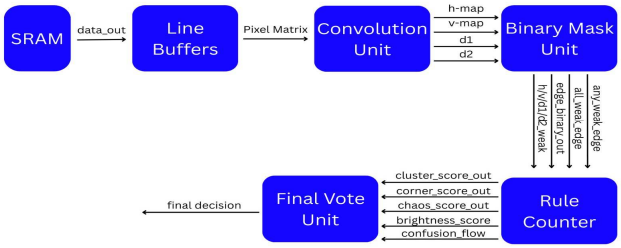


Figure 1 – Block Diagram for Proposed Architecture

B. Window Formation and Line Buffering

The input image is stored in on-chip SRAM and streamed at one pixel per cycle. Two line buffers and shift registers generate a 3×3 sliding window, enabling continuous spatial processing without pipeline stalls.

C. Gradient Computation Using Scharr Operator

A Scharr-based compute unit extracts gradients in four directions: horizontal (H), vertical (V), and diagonals (D1, D2). The Scharr operator is chosen for its superior rotational symmetry and edge sensitivity. The outputs include signed gradient maps (h_map, v_map, d1_map, d2_map).

D. *Adaptive Binary Mask Generation*

Gradient magnitudes are compared against programmable thresholds to produce weak and strong edge masks. Additional masks include cluster (OR of weak edges), chaos (AND of weak edges), and corner (intersection of strong H and V edges), forming inputs for rule evaluation.

E. Rule Counter and Score Accumulation

A rule counter accumulates frame-level metrics: cluster score (weak edges), corner score (strong intersections), chaos score (all weak edges), and brightness score (gradient magnitude). A confusion flag is raised under excessive chaotic edge patterns. Counters reset every frame.

F. Final Diagnosis via Voting Logic

A voting unit compares scores with predefined thresholds and assigns votes per rule. If the total votes exceed a threshold (e.g., ⩾3), the output is classified as a fracture. This ensures deterministic and interpretable system operation without relying on machine learning.

IV. RESULTS AND DISCUSSION

A. MATLAB-Based Functional Validation

The proposed algorithm was first validated using a MATLAB model. For normal X-ray images, smooth gradients resulted in low rule scores and correct classification. In fractured cases, strong discontinuities increased cluster and chaos activations, leading to correct detection. These results confirm functional correctness prior to hardware implementation.

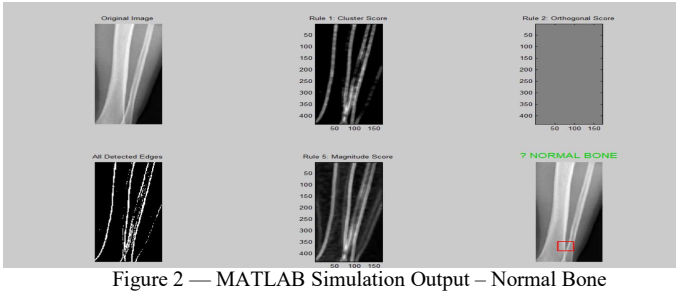


Figure 2 — MATLAB Simulation Output – Normal Bone

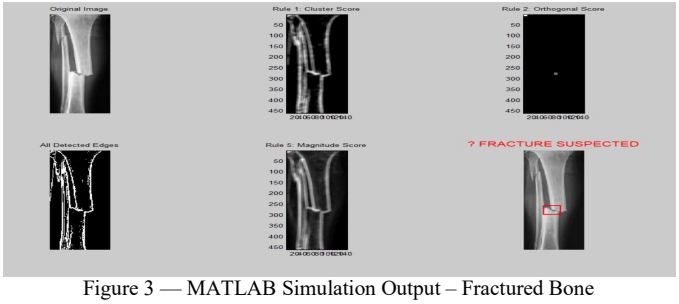


Figure 3 — MATLAB Simulation Output – Fractured Bone

B. Simulation Results

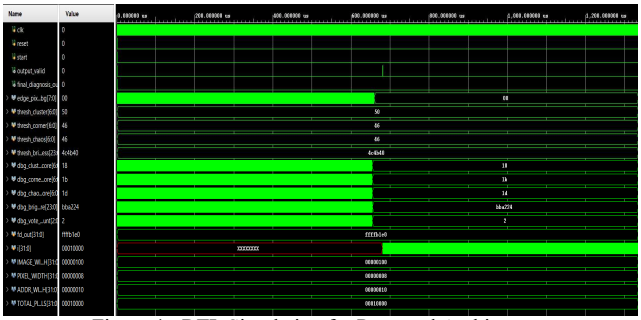


Figure 4 - RTL Simulation for Proposed Architecture

RTL simulation verified correct datapath and control operation, including pixel streaming, rule accumulation, and decision logic. The voting unit produced correct frame-level outputs with proper synchronisation, confirming accurate hardware realisation of the rule-based approach.

C. Synthesis Results

The architecture was initially synthesised using Cadence GENUS with a 90-nm CMOS standard-cell library for RTL validation and functional verification. Final physical implementation and post-layout optimisation were performed using a 45-nm CMOS technology library.

1) Fracture vs. Normal Output from RTL Simulation

A. Case Study 1 – Fractured Bone

Table 1 — Rule-Based Scores (Fractured Bone)

Metric	Value
Cluster Score	120
Corner Score	39
Chaos Score	99
Brightness Score	5,930,506
Vote Count	4
Final Decision	**1 (Fractured)**

B. Case Study 2 – Normal Bone

Table 2 — Rule-Based Scores (Normal Bone)

Metric	Value
Cluster Score	24
Corner Score	27
Chaos Score	29
Brightness Score	12,296,740
Vote Count	2
Final Decision	**0 (Not Fractured)**

C. Overall System Accuracy

The proposed ASIC was evaluated on 46 X-ray images (29 fractured, 17 normal). The system correctly classified 39 cases, achieving ~85% accuracy. Misclassifications mainly occurred in low-contrast and subtle fracture cases.

These results demonstrate that gradient-based features with deterministic voting provide reliable and interpretable fracture detection, making the design suitable for real-time embedded medical applications.

2) Area Analysis

Post-synthesis area is 91,866 µm² with 7,731 standard cells (4,266 sequential, 3,042 combinational). The dominance of sequential elements is due to line buffers and rule counters, confirming suitability for compact embedded implementation.

Table 3 — Post-Synthesis Area Summary

Metric	Value
Total Area	**91,866.467 µm²**
Total Standard Cells	7,731
Sequential Cells	4,266
Combinational Cells	3,042
Inverters & Buffers	422

3) Power Analysis

Total power consumption is 5.729 mW, with internal power dominating (5.005 mW), while switching and leakage remain minimal. This indicates good energy efficiency of the design at the synthesis level.

Table 4 — Post-Synthesis Power Summary

Power Component	Value(mW)
Internal Power	5.005
Switching Power	0.216
Leakage Power	0.528
Total Power	**5.729**

C. Physical Design Results (Post-Layout, Innovus)

The design was implemented using Cadence Innovus, including floorplanning, placement, CTS, routing, and post-route optimisation. DRC and LVS were successfully cleared, ensuring manufacturability.

1) Post-Layout View

The final layout shows uniform cell distribution and well-managed routing without congestion, confirming a robust physical design.

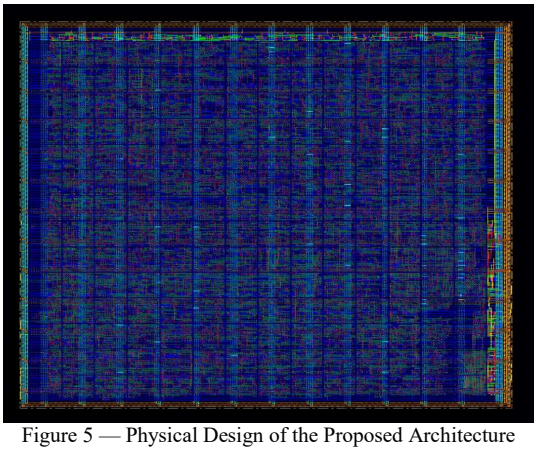


Figure 5 — Physical Design of the Proposed Architecture

2) Post-Layout Area

The final post-layout area obtained after placement and routing is 33,047.46 µm² (80% utilisation) due to routing and placement overheads in 45-nm technology.

Table 5— Post-Optimisation Area Summary

Metric	Value
Total Area	**33,047.460 µm²**
Sequential Cells	7554
Combinational Cells	Included in total; library does not separate
Total Standard Cells	7554
Utilization	80%

3) Post-Layout Power

Total power is 1.151 mW (internal: 0.776 mW, switching: 0.3715 mW, leakage: 0.003 mW), with internal power dominating due to sequential activity.

Table 6 — Post-Route Power Summary

Power Component	Value(mW)
Internal Power	0.776
Switching Power	0.3715
Leakage Power	0.003326
Total Power	**1.151**

4) Operating Frequency

The design operates at 77 MHz with positive timing slack and no violations, meeting real-time processing requirements.

V. CONCLUSION

This work presents a complete ASIC-based fracture detection system using a fully hardware-optimised pipeline. The architecture integrates Scharr gradient extraction, adaptive masking, rule-based scoring, and voting logic, eliminating the need for machine-learning models. Implemented in 45-nm CMOS, it operates at 77 MHz with 1.151 mW power and 33,047.46 μm² area, demonstrating a compact and energy-efficient design.

Post-layout results confirm deterministic operation and real-time performance, achieving ≈85% classification accuracy on a dataset of 46 X-ray images. The low power, small area, and predictable latency make the system well-suited for portable and embedded medical devices.

Future work includes extending support for higher-resolution images, incorporating on-chip noise reduction, and exploring hybrid rule-learning approaches to improve accuracy.

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